

Sequential Pareto-Optimal Decisions Made During Emergent Complex Systems of Systems: An Application to the FAA NextGen

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ABSTRACT

This article addresses the challenges facing participants, including stakeholders, decision makers, systems engineers, and risk analysts in modeling and managing the development of on-going emergent complex systems of systems (S-o-S), with a focus on the centrality of the state variables and the time frame. More specifically, this article addresses the critical role that shared (common) states and decisions, and the time frame play in modeling the interconnectedness and interdependencies among the subsystems that constitute emergent (S-o-S). The theoretical and methodological concepts are harmonized through their relevance to the ongoing emergent Federal Aviation Administration's (FAA's) Next Generation (NextGen) complex (S-o-S) project. Appropriate parts of the Dynamic Roadmap for risk modeling, assessment, management, and communication and their companion principles and guidelines, are used to guide the analyses. Moreover, decisions made at time t for one subsystem will subsequently change the states of that subsystem as well as affect other interconnected and interdependent subsystems that share states. This phenomenon is profoundly important even when the time frame of the affected states is measured in months or years, as is the case in the application discussed in this article. The centrality of the time frame in decision making is even more important when the impact of current decisions on future options affect not only the intended subsystem, but can also inadvertently affect other subsystems in the future. A Flowchart relates the (S-o-S)-based theory and methodology to the FAA's NextGen complex (S-o-S) project. © 2014 Wiley Periodicals, Inc. Syst Eng 18: 28–44, 2015

Key words: systems of systems; sequential decisions; NextGen; Pareto-optimal policies; unmanned aerial vehicle (UAV)

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Preface

Addressing the third question in risk management, “What are the impacts of current decisions on future options?” is critically important for emergent complex (systems of systems [S-o-S]), because their conception, the evolution of their requirements and specifications, their design and development, and their ultimate operation can span several years. Furthermore, the sequential decisions made during the development of each subsystem of these complex (S-o-S)

will most likely affect the development of other subsystems in the future with the expectation that, in their totality, all subsystems are ultimately expected to operate as an integrated harmonious (S-o-S). For example, decisions to achieve specific outcomes made on subsystem A of an emergent (S-o-S) can change the states of that subsystem as well as affect other interconnected and interdependent subsystems that share states with subsystem A.

The risks and challenges associated with this sequential decision making process are of paramount importance for the emergent (S-o-S) currently under development by the U.S. Federal Aviation Administration (FAA) [2012], known as Next Generation (NextGen). The interconnectedness and interdependencies of these emergent (S-o-S) are modeled and represented through shared (common) states and decision variables existing among the subsystems, as advanced by the Phantom System Models [Haimes, 2012a]. To avoid abstraction, and to more clearly communicate the concepts and methodologies introduced in this article, appropriate examples of the shared states and decisions are linked to NextGen to demonstrate their relevance and practicality. In this sense, relating the systems-based methodological approach and its application to the NextGen enterprise becomes one integral theme. The technical presentation is described through the current and on-going modernization of the National Airspace System (NAS), one of the most all-encompassing projects ever undertaken by the FAA.

The following premises guide the technical presentation in this article, given that the time frame for the development of NextGen spans two decades:

- i. Sequential decisions made during the emergence of complex (S-o-S) must be dynamically Pareto-optimal, [Pareto, 1896] and can be generated using the Envelope Approach [Li and Haimes, 1987; Haimes 2009b]. The criticality of the time frame, within which the ultimate decisions are most likely to accomplish their intended objectives, and the multiple decision makers and stakeholders, will require agreed upon policies commensurate with the satisficing principle introduced by Simon [1956].
- ii. The emergent behavior of (S-o-S) arises from the various tradeoffs made by multiple stakeholders who have multiple goals and objectives affecting future decisions and outcomes.
- iii. The desired emergent behavior of S-o-S, as a whole, represents a collective objective driven by separate subsystem level decisions made by individual stakeholders based on “local subsystem objectives.” Note that the motivation for changes in a subsystem can be twofold: (a) the objectives of the local subsystem and (b) the larger desired emergent behavior for the (S-o-S) as a whole. The FAA is generally interested in the desired emergent behavior of the overall enterprise in addition to the local effect. DeLaurentis [2005] was a pioneer in addressing the air transportation as S-o-S.
- iv. The collaboration among the stakeholders regarding tradeoffs associated with each objective is essential to the optimal sequencing of decisions and to making enterprise strategy-level decisions.

Furthermore, the appropriate elements of the systems-based guiding principles for risk analysis [Haimes, 2012b], and the related Dynamic Roadmap, will be applied to the NextGen enterprise. The challenges associated with the modeling and development of emergent (S-o-S) are highlighted, with a focus on the centrality of the state variables and time frame. One theme that transcends the entire discussion in this article is the critical role that shared (common) states, decisions, and the time frame play in modeling the interconnectedness and interdependencies among the subsystems that constitute emergent S-o-S. The theoretical and methodological concepts are harmonized through their relevance to ongoing emergent complex S-o-S in the FAA’s NextGen program.

1. METHODOLOGICAL APPROACH

1.1. Introduction

The increasing complexity of shared infrastructure projects that span the public and private sectors have begun to exceed the capabilities of traditional management techniques to plan, construct, and control them. Every new generation of technology and operational practice increases in complexity and ambition. This trend has led to an explosion in the sophistication of government and private industry projects designed to develop and/or acquire technology-based systems. Unpredictable interactions and interdependencies among individual and community users, software–hardware, and organizations commonly characterize these systems. This complexity has imposed a complementary rise in the level of “unpredictable” adverse consequences, in terms of technological risk (not meeting a project’s performance criteria, e.g., failing to deliver expected benefits for individual investors or the desired behavior of the emergent system-of-systems [S-o-S] as a whole), and programmatic risk (cost overrun, time delay in project completion).

In general, there is no one right way to perform the complex process of risk modeling, assessment, management, and communication associated with small or large projects. Often the best approach is driven by the unique characteristics of the system under consideration. *Regardless, there are certain principles and guidelines, introduced in this article, that apply universally to emergent complex S-o-S.*

1.2. The Complexity of Planned and Emergent S-o-S

Emergent complex S-o-S, are by definition composed of several subsystems, and are commonly in a constant state of flux, modification, and readjustment caused by the fluctuations in one or more subsystems. Most notably, these seemingly random driving forces, denoted by the “evolving base” are dynamically shifting interdependencies and realities based on changing [Haimes, 2012b]: (i) goals and objectives; (ii) stakeholders, decision makers, and interest groups; (iii) organizational, political, and budgetary considerations; (iv) reorganization and reallocation of key personnel; and (v) requirements, specifications, delivery dates, users, and clients.

The literature is replete with definitions of what constitute S-o-S, albeit not how to model them. Maier [1998] identifies five characteristics of S-o-S:

(i) Operational independence of the individual systems; (ii) managerial independence of the systems; (iii) geographic distribution; (iv) emergent behavior; and (v) evolutionary development.

Other contributors to the literature on complex S-o-S include Bar-Yam [2003a, 2003b], Blaumberg [1977], Eisner [1993], Jamshidi [2009a, 2009b], and Haimes [2007, 2012a]. Sage and Cuppan [2001] provide the following representative definition of emergent behavior in the context of a system of systems:

The S-o-S performs functions and carries out purposes that do not reside in any component system. These behaviors are emergent properties of the entire system of systems and not the behavior of any component system. The principal purposes supporting engineering of these systems are fulfilled by these emergent behaviors. (p. 326)

One of the major challenges facing professionals in charge of the multiple stages in the development of complex S-o-S, such as the FAA's NextGen (to be subsequently introduced), is harmonizing the myriad sequential decisions, whether regarding policies and procedures or structural issues, so that decisions made at one point will not conflict with, or adversely affect, future options. By their nature, emergent S-o-S are composed of multiple interconnected and intra- and interdependent subsystems, each with a unique mission, and human, cultural, organizational, geographical, and technological characteristics that span cyber and physical infrastructures. In addition, a hierarchy of diverse stakeholders and decision makers, multiple noncommensurate and often conflicting and competing goals and objectives that span multiple time frames, and assorted policies and procedures serving each subsystem are features of these systems. Clearly, it is essential to coordinate the planning, design, specification, and construction of each subsystem, to develop appropriate operational policies and procedures that meet the needs of each subsystem, and to ensure a harmonious integration of the subsystems with each other. This constitutes a challenging process of risk modeling, assessment, management, and communication. The common denominators that both transcend and link the above activities are the *time frame* and the *shared states* that constrain as well as enable the subsystems to function, to the extent possible, as a harmonious S-o-S.

One of the challenges facing the developers of emergent S-o-S is to navigate the sequential temporal decision-making process while being cognizant of the dynamic multiobjective tradeoffs leading to a satisficing set of policies that are responsive to the broader goals and objectives of emergent S-o-S enterprise. This process is also aimed at harmonizing the myriad decisions associated with each emerging subsystem and within emergent S-o-S.

In sum, the development of a sequential multiobjective decision-making process and its deployment on the NextGen system of systems would demonstrate that decisions made, policies formulated, and procedures and regulations promulgated during one period would not cause future detrimental technical and programmatic risks to the subsystems within NextGen, nor would they adversely affect or impact the

multiple objectives and stakeholders associated with the NextGen S-o-S.

1.3. The FAA NextGen System of Systems

Applications presented in this article are drawn from the FAA NextGen multiyear, multiagency, and multibillion dollar project. NextGen is a project to modernize the National Airspace System (NAS). Decisions for this complex enterprise are made by a combination of individual stakeholders and organizations in order to achieve specific objectives and outcomes. At the same time, the FAA has a desired set of emergent behaviors for the evolution of the NAS, which will be the product of these many separate decisions. NextGen epitomizes interconnected and interdependent complex S-o-S.

NextGen is a public-private endeavor involving over 30 major systems and 190 government subsystems, each of which has its own platform and many stakeholders. The private sector consists of hundreds of commercial aircraft operators with hundreds of models of aircraft, and thousands of private aircraft operators. These aircraft operate out of over 5000 airports, all with local management. All of these components are part of the NextGen emergent system of systems. From the FAA perspective as a regulator and service provider, the decisions for FAA components must serve the diverse population of private sector stakeholders. An aim of NextGen is to create incentives for all participants to invest in capabilities that will contribute to the desired emergent behaviors of safety, flight efficiency, capacity, and the environmental impact of the S-o-S as a whole.

FAA's NextGen is being planned with three major 5-year implementation windows, known as Alpha (2010–2015), Bravo (2015–2020), and Charlie (2020–2025). An inherent characteristic of an emergent system of systems such as NextGen is the long time frame from its development to its realization. Components, such as aircraft and airport modifications will be expected to remain in service for 20 years or more. Each decision, therefore, will have long lasting consequences. Furthermore, the timing of changes must be planned and aligned with changes in other subsystems that share common states. In particular, the coordination of the planned services will require changes in ground systems, and synchronizing of aircraft and flight crews.

1.4. Three Major Sources of Risk that Threaten Emergent Interdependent S-o-S Characterized by Intrinsic Shared States

Technical, programmatic, and enterprise performance risks are all concerns of emergent S-o-S. The first two risks apply to the subsystem level, while enterprise risk is the unexpected consequence in the desired emergent behavior of the S-o-S as a whole. More specifically:

- i. *Technical risk* occurs when a project fails to meet its performance criteria. This includes hardware and software failures and requirement shortfalls. Technical risk can

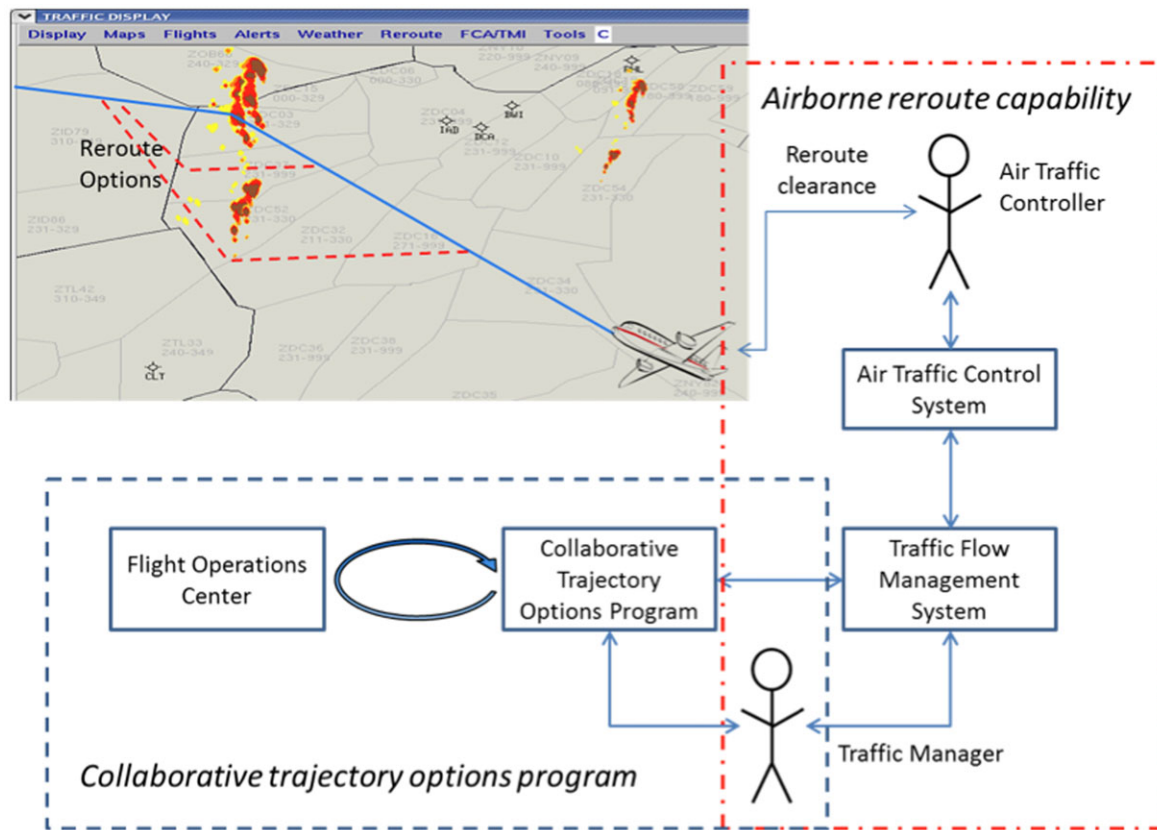


Figure 1. Airborne Reroute and Airborne Trajectory Options Capabilities.

occur when an emergent forced change causes significant changes in the behavior of one or more subsystems, leading to a cascading failure of the S-o-S. The phrase “emergent forced changes” connotes *trends in external or internal sources of system risk that may adversely affect specific states of that system*.

- ii. *Programmatic risk* denotes cost overruns and delays in the schedule. This source of risk characterizes many emergent large-scale systems, especially for subsystems with significant software components.
- iii. *Enterprise performance risk* leads to negative outcomes due to changes in the system. When the performance of an *emergent* S-o-S does not meet the collective desired behavior, whether by unanticipated, undetected, misunderstood or ignored emergent forced changes from within or from outside a system, the state changes may result in a negative emergent behavior. Therefore, it is imperative to be able, through scenario structuring, modeling, and risk analysis to envision, discover, and track emergent forced changes [Haimes, 1981, 2009b] and to understand the likely consequences for emergent behavior. The interdependency among sub S-o-S is a manifestation of the intrinsic shared states among them. Consequently, emergent forced changes that affect intrinsic shared states of one subsystem will affect all other subsystems that share the same intrinsic states. In sum, the essence of interdependencies among sub S-o-S is grounded on the shared states among the subsystems.

Consider the case where a decision maker acts believing that a subsystem change will create a desired behavior; however, the change is actually insufficient to provide the desired performance outcome. In a hypothetical example from NextGen, consider two capabilities that may each be necessary, yet not sufficient to achieve the desired benefit. The first is an *airborne reroute* path dependent *capability*, which replaces a series of paper note exchanges and phone calls with electronic exchange of information between multiple parties needed to direct an aircraft to an alternate route, as depicted in Figure 1. The second capability, *collaborative trajectory options program*, also shown in Figure 1, enables the aircraft operator flight operations center to indicate a preference for airborne trajectory options. Each of these capabilities are necessary to create more flexible routing around weather, but are individually insufficient to do so. It may be that without including both capabilities, the use of airborne reroutes will fail to achieve the desired performance. This could be anticipated by examining the shared states for the two capabilities. Both capabilities involve knowledge of the airborne flight trajectory, procedures for avoiding weather, and operating environments involving weather. The implication is that all capabilities with shared states must be evaluated in collaboration with the independent decision makers for the necessary and sufficient conditions to create a desired emergent behavior. Such decisions become corequisite.

In other cases, decisions may transfer program- or technical-risk among decision makers. This would be the case

if one of the capabilities in the example above, either the airborne reroute capability or the airborne trajectory options capability, were implemented with specifications that reduced the risk of one by adding program risk to the other. If coordinated as part of the overall design of a combined capability then this risk management should lower the overall risk. If this transfer of risk occurs as a result of separate, uncoordinated program designed activities, then this represents a risk transfer from one decision maker to another, which ultimately can degrade both the desired performance of the S-o-S as a whole and the level of trust. A key concept in managing the risk of not achieving the desired emergent behavior is that *the critical decisions that warrant additional collaboration among decision makers can be identified through the analysis of the shared states and can be addressed proactively*. Risk mitigation should include coordination of the sequence and scope of the individual decisions with knowledge of the implications on other decisions.

Emergent forced changes are tied to risk management and they cannot be overlooked. Because of their broad consequences to the contractor, the customer, the user, the analyst, and other stakeholders, numerous parties have a stake in the assessment and management of the above sources of risk. To effectively manage these sources of risk, these parties must meet collectively and periodically to ensure that proper risk management is being conducted. No party can be ignored because each brings a different perspective and background knowledge.

1.5. Criticality of the Time Frame in Sequential Decision Making

One of the characteristics of emergent S-o-S is the long time frame associated with their development and realization. The initiation of a new system starts with defining mission needs and conducting analysis. Such activities can span two decades for completion and would likely be impacted by emergent forced changes that must be addressed. As noted earlier, the planning time frame for the FAA NextGen project is segmented into three major design and development periods of five years each, known as Alpha, Bravo, and Charlie. Indeed, decisions made at any time period are “path dependence.” Liebowitz and Margolis [2000] eloquently elaborate on the importance of the time frame concept: “Most generally, path dependence means that where we go next depends not only on where we are now, but also upon where we have been.” Decisions made during any period can have significant consequences on the state of the system both at the time decisions are made and during subsequent periods; namely, we have path-dependence decisions. By altering the states of the S-o-S, each decision may adversely affect the size, location, functionality, and schedule of future changes. In addition, the large number of major structural and policy decisions exacerbate tracking such forced changes.

Consider, as an example, the following two decisions made in parallel. Decision *A* will be an FAA decision to introduce schedule-based merges within a terminal area. Referred to as terminal sequencing and scheduling, this is a flow management capability in which specific times and tolerances are used to smoothly interleave two flows of traffic. Deci-

sion *B* will be a decision to structure the arrival routes in a way that improves the ability to model and execute flight approaches to tighter tolerances. If decision *A* were made first it would provide insights concerning objectives for decision *B*. However decision *A* may potentially set tolerances that are unachievable by the models or execution methods available at the terminal. If decision *B* were made first, it would establish the flows that would be criteria as to whether *A* remains a viable option for that terminal. These policy decisions are of nontransitive properties—they must be followed not only at the right time but also in the right sequence. Decision pairs with this nontransitive property may be identified by analysis of decisions that share common states in the system. Note that decisions *A* and *B* affect the following three shared states: (i) state of regulations; (ii) state of operating environment, and (iii) state of specific technology.

1.6. Dynamic Multi-Objective Decision Making

Most decisions involve implicit or explicit dynamic tradeoffs among multiple, often conflicting and noncommensurate goals and objectives. *Indeed, analyzing and understanding the impacts of decisions on multiple objectives and their associated tradeoffs that affect different time frames constitute a foundational maxim in prudent decision making.*

Given the often conflicting, competing, and noncommensurate nature of the multiple objectives associated with NextGen, and because the tradeoffs among them are complex from multiple perspectives, it is impractical to consider assigning them weights to use in making meaningful decisions. The decision-making process must avoid an overly constrained feasible set for which there is no answer. In this article, we adopt the concept of satisficing originally introduced by Herbert Simon [1956], which builds on an even earlier paper by him [Simon 1947].

Any attempt to “optimize” a set of objectives, even within the Pareto-optimal sense, that are limited to present needs and aspirations, or that are not responsive to emergent or future forced changes, could lead to severe unintended consequences. (Informally, *a solution to two-objective functions is termed Pareto-optimal if improving one objective function can be achieved only by degrading the other* [Intrilligator 1970].) In other words, given the nature of the evolving, adaptive, incremental, and multistage S-o-S project, present decisions and policies must take into consideration that emergent forced changes may produce catastrophic or irreversible consequences on the project in the future. Thus, a systemic approach to addressing the multistage decisions during Alpha, Bravo, and Charlie periods should (i) continuously assess and evaluate precursors to potential emergent forced changes (building on the theory of scenario structuring), (ii) balance present multiple objectives with potential or perceived emergent future needs and objectives, and (iii) add more flexibility to present policy formulation to ensure against adverse emergent or unintended catastrophic consequences.

Robustness stems from evaluating the consequences and future flexibility of two preferred Pareto-optimal policies where potential future environments would yield a distinct choice between two options presently perceived as

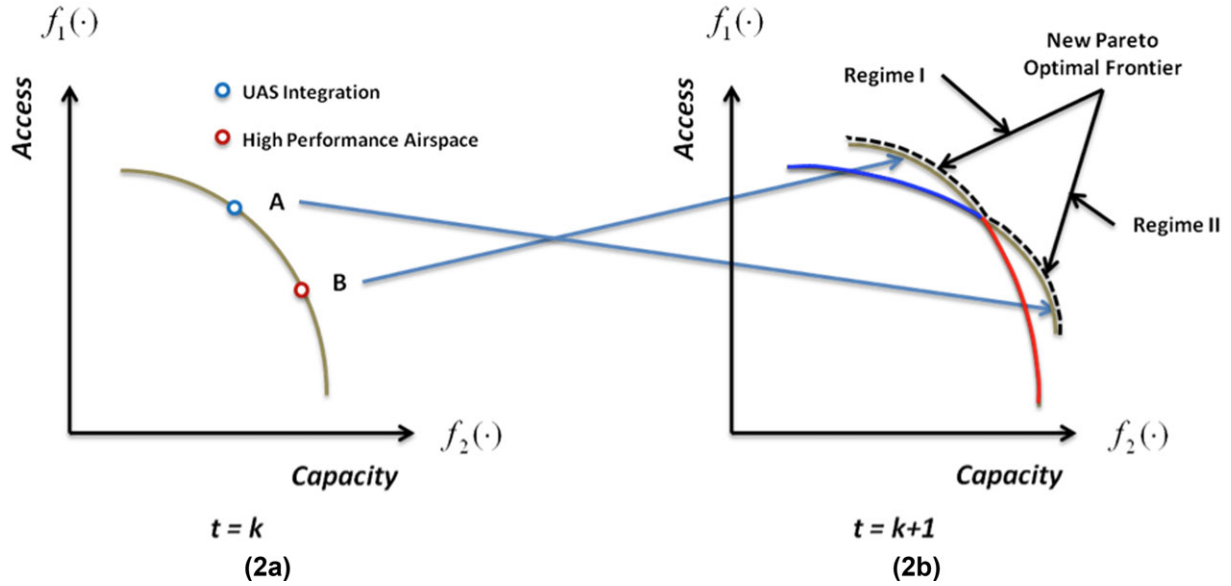


Figure 2. A dynamic Pareto-optimal frontier is related for two objective functions: Access and Capacity.

seemingly equivalent. Assessing the temporal impacts on future multiobjective-based options is especially critical for rapidly changing technological systems, such as NextGen. The *Envelope Approach*, which enables dynamic multiobjective tradeoff analyses [Li and Haimes [1987], will be introduced below.

Furthermore, dynamic systems such as NextGen, where the time frame is of paramount importance, introduce another challenge to both modelers and decision makers. Namely, ignoring or not accounting for the changes in the states of a safety-critical dynamic system as time progresses (e.g., such as the integration of the Unmanned Aircraft System (UAS) within the NAS), would render static-based decisions misleading, if not disastrous. Recognizing this fact, methods such as event trees, decision trees, and process control can account for the impact of current decisions on future options [Werner, Haimes, and Anderegg, 2014].

Consider the following two objectives at time k depicted in Figure 2: Access, $f_1(\cdot)$, in the Y-axes representing increased flight capacity resulting from NextGen, and Capacity, $f_2(\cdot)$, in the X-axes representing improving flight efficiency at time k . Figure 2a presents two Pareto-optimal policies **A** and **B** selected at time $t = k$. When the projections of policies **A** and **B** are plotted for period $t = k+1$ on in Figure 2b, they reveal that only the “Envelope” depicted in the dotted line of the intersection of the two Pareto-optimal frontiers constitutes the new Pareto-optimal frontier (for $t = k+1$). Thus, given the availability of a dynamic model of the two-objectives, each policy **A** (associated with UAS integration) and **B** (associated with high performance airspace) at time $t = k$ (see Fig. 2a) would generate future (new “Envelope”) Pareto-optimal frontier at $t = k+1$ (see Fig. 2b). In sum, although at time $t = k$ policies **A** and **B** are equivalent in the sense of Pareto optimality, their projected consequences at time $k+1$ as depicted in Figure 2b in the new Pareto-optimal frontier are not. Consider Efficiency, $f_3(\cdot)$, (of the NAS) as a third objective of the NextGen as depicted in Figures 3a and 3b:

(i) at $t = k$, the decision makers desire to maximize Access to and Capacity of the NAS at time $t = k$ (Fig. 3a); However, at the next period, $t = k+1$, the decision makers would want to consider a third objective; namely, (ii) maximize Access to and Efficiency of the NAS. Policy (Fig. 3b). Similar to Figure 2, policy **A** at time $t = k$ would be inferior at $t = k+1$ for Regime I in Figure 3b; and policy **B** at time $t = k$ would be inferior at $t = k+1$ for Regime II in Figure 3b. In other words, given the “Evolving Base” and the expected changes in circumstances and priorities, decision makers and policy analysts can use the dynamic multiobjective analysis, using the “Envelope” approach to project scenarios, given the inevitable evolving base. More will be discussed in the next section. The Envelope Approach was developed to account for the propagation of Pareto-optimal solutions from one period to another [Li and Haimes, 1987].

Figure 4 presents a final example of dynamic Pareto-optimal frontier to n periods through the Envelope Approach. Note that the solid-line’s outer curve represents the envelope of the combined Pareto-optimal frontier of policies **A**, **B**, **C**, and **D** for the $k+1$ st period. In sum, the value of the dynamic Pareto-optimal frontier using the envelope approach can be explained in the following more generalized example: All four decisions in Figure 4 are equivalent in the sense of Pareto-optimality, however if we project those decisions to time $t = k+1$, each of the policies **A** through **D** would project a different Pareto-optimal frontier, the envelope of which is the only Pareto-optimal frontier for $t = k+1$. More specifically, consider the case where policies **A** through **D** represent during time $t = k$ distinct means to deliver air traffic control services at nontowered airports with (i) different cost effectiveness, and (ii) different airport throughput implications. Given future operating environments in period $t = k+1$, the introduction of different classes of UAVs would produce different Pareto-optimal frontiers associated with cost-effectiveness and throughput for each option.

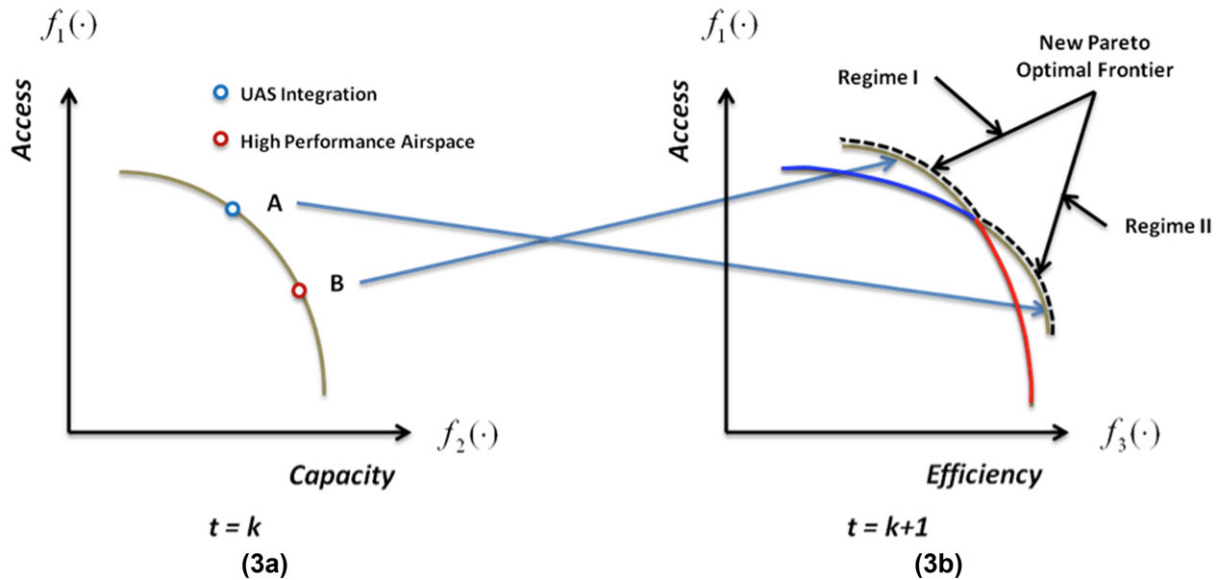


Figure 3. A dynamic Pareto-optimal frontier is related for three objective functions: Access versus Capacity and Access versus Efficiency.

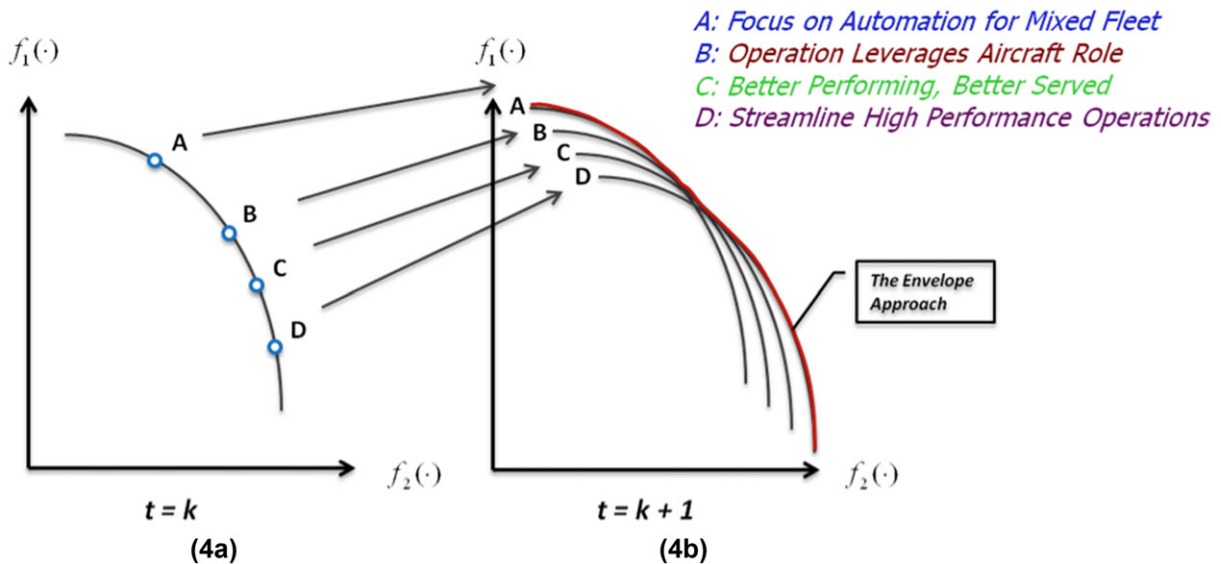


Figure 4. The envelope of the combined Pareto-optimal frontier of policies A, B, C, and D for $k + 1$ st period.

The future environment would likely be similarly affected by other changes in period $t = k+1$, such as the use of data communications between pilot and controller, the expansion of surveillance capabilities, changes in the high-altitude airspace flows and allocation of controller workforce across facilities driven by demand.

1.7. Judgment and Values in Decision Making

In every decision process there are factual elements and value elements. Factual elements are those that can be verified scientifically and subjected to scientific machinery and are likely to lead to other verifiable elements. Value elements, on the other hand, defy all forms of scientific verification and

treatment. A collection of value elements and their sources constitutes a value system. Judgment, which signifies the act of giving an opinion, is the most common value element in any decision-making process [Chankong and Haimes, 1983, 2008]. The decision-making process is often an amalgamation of real or perceived facts and of value judgment.

The development of the emergent NextGen S-o-S is necessarily driven by both fact-based and judgmental decisions. For the publicly funded and operated NextGen complex S-o-S, analytically harmonizing logical inputs with the political decision-making process is often not a smooth process. In such cases, past experiences can skew the sense of where investments are needed to properly assess and manage technological or programmatic risks. Similarly, biases affect the

relative value of desired behaviors and outcomes in a multiobjective decision-making environment.

In this article, we posit that the amalgamation of facts and value judgments has an important role to play in the complex dynamic multiobjective decision-making process, where judgment plays an integral role. Modeling and managing NextGen as a S-o-S enable stakeholders and decision makers to analyze, model, and measure desired emergent behaviors' performance of the S-o-S as a whole. This dynamic multiobjective process constitutes a nontrivial challenge to the community that must decide on acceptable tradeoffs among dynamic, noncommensurate, competing, and conflicting objectives that characterize the overall performance of NextGen.

Similar to the existence of a hierarchy of sub- and sub-substates of a system, so too are there subobjectives, and sub-subobjectives, especially for emergent complex S-o-S. The highest level of this structure generally represents the broad overall objectives that are instrumental in initiating the multiobjective decision problem in the first place. The overall objectives of NextGen, safety, efficiency, capacity, and environment, are a case in point. However, these objectives are often vaguely stated and, hence, not operational. For example, the NextGen project must minimize inefficiencies for flight operators, while also holding the accident rate to 10^{-6} and simultaneously increasing the national airspace capacity and ensuring stewardship of the environment. On the other hand, farther down the hierarchy the objectives are more specific and more operational than those at the higher levels, and they are at least perceived as a means to achieve the higher ends represented by the objectives at the higher level. As a result, objectives at the lowest level of the hierarchy are the most specific and the most operational.

For example, in the NextGen system of systems, simultaneously reducing excess taxi time on the ground in a way that reduces the risk of runway incursions represents a win-win—two synergistic outcomes with a positive change. For example, Figure 3 represents a dynamic Pareto-optimal frontier for two objectives: access versus capacity and efficiency. From a capacity perspective, the two capabilities have a clear tradeoff with access, but when considering efficiency the relative tradeoff may reverse, then the criteria for the judgment call become important.

Consider the introduction of procedures to integrate UAS within the NAS, or alternatively to create performance-based airspace as a means of accommodation of the UAS. The tradeoffs could be quantified, but the acceptable policy on the Pareto-optimal frontier would be a judgment call.

These judgment calls are made more complex by the combination of tradeoffs among shared states highlighted in Table I. For example, three NextGen decisions that fit in such a cluster of shared state variables, include: (i) a decision to create high performance airspace (segregating certain operations); (ii) a decision on automation needs to serve a more diverse fleet in terms of aircraft and navigation performance; and (iii) a decision on the aircraft role in separation and collision avoidance. Each of these decisions has a different effect on the three objectives: access, capacity, and efficiency. The criteria or basis for judgment in balancing between these objectives must be clear to the analyst who must study the

three decisions as a set. The resulting dynamic multiobjective tradeoffs must be represented in a three dimensional space.

The U.S. Congress mandated that the FAA develop a plan for the integration of UAS within the NAS within the NextGen enterprise. The known and precursors to unknown multiple sources of risk associated with such integration must be carefully assessed and addressed. Balancing the tradeoffs among the multiple competing goals and objectives aspired by the many stakeholders and decision makers constitutes a daunting task that can benefit from the dynamic analyses depicted in Figures 3 and 4. These trades may be informed by fact, but are still a judgment call in the end, and they constitute some of the drivers that influence such tradeoffs.

The very nature of emergent S-o-S is that multiple decision makers direct the evolution of the enterprise. Decisions made by an individual stakeholder reflecting local objectives of one subsystem might be perceived as isolated, but it will likely affect—positively or negatively—other public decisions made to incentivize a desired emergent behavior. Consider NextGen's efforts to increase access to runways in marginal meteorological conditions. A capability using improved navigation procedures applicable to all trained aircraft operators would raise the total throughput for the airport by mitigating the loss of parallel runway throughput. An alternative solution uses enhanced visual systems to deliver benefit in the form of greater access as reduced levels of visibility directly to flights that are so equipped and trained. The overall capacity of the airport in reduced visibility is not affected by the second solution. Both decisions impact on the local objectives, but they require different investments by aircraft operators and result in different emergent behavior objectives. In the first case all aircraft operators share in the benefit from the investment in training the flight crews regardless of who invests, while in the second case those who invest in the visual system have more exclusive rights to the benefit as only they will receive improved service (a differentiated value for the investor).

2. PRINCIPLES AND GUIDELINES

This section presents definitions, principles, and premises for the planning, deployment, and management of emergent S-o-S.

2.1. Centrality of the States of a System:

Given a system's model, the smallest set of independent system variables such that the values of the members of the set at time t_0 along with known inputs, decisions, random, and exogenous variables determine the value of all system variables for all $t > t_0$. The shared states among the subsystems that constitute the NextGen S-o-S play a major role in modeling, understanding, and ultimately harmonizing (to the extent possible) the myriad time-critical structural and nonstructural policy decisions that are made by the multiple stakeholders engaged in this complex process. Table I presents shared states among the major NextGen objectives. These shared states play a central role in highlighting and

Table I. Shared States Among the Major NextGen Objectives.

Efficiency	Safety	Capacity	Environment
1. ATM Technology	1. ATM Technology	1. ATM Technology	
2. Operating Environment	2. Operating Environment	2. Operating Environment	2. Operating Environment
3. Personnel & Operations	3. Personnel & Operations	3. Personnel & Operations	
4. Aircraft Responsiveness	4. Aircraft Responsiveness	4. Aircraft Responsiveness	
5. Information Quality	5. Information Quality		
7. Culture	7. Culture	7. Culture	
8. Budget		8. Budget	8. Budget
9. Reliability (6 Sigma)	9. Reliability (6 Sigma)	9. Reliability (6 Sigma)	9. Reliability (6 Sigma)
10. Trust	10. Trust	10. Trust	
11. Standardization	11. Standardization		
12. Congestion	12. Congestion		
	13. Product Reliability		
21. Runway Usage		21. Runway Usage	21. Runway Usage
23. Fuel Usage			23. Fuel Usage
			24. Noise

appreciating the complexities in the decision-making process that are caused by the interconnectedness and interdependencies among the subsystems, by the objectives, and by the corresponding stakeholders and decision makers. Furthermore, the shared states streamline the essential harmonization process associated with reaching acceptable tradeoffs among the dynamic multiple objectives as well as among the principal stakeholders and decision makers, and, ultimately, acting upon those tradeoffs.

The complexity of the quantification of the multidimensional risk function has been modeled by relying on the states of the system. Indeed, the multidimensional probabilistic consequences, resulting from an initiating event, yield a multidimensional risk function whose modeling and quantification can be achieved most effectively through the states of the affected system. Furthermore, both the vulnerability and the resilience of a system to any specific initiating event are manifestation of the states (e.g., *physical, technical, organizational, and cultural*) of the affected system. Thus, the consequences resulting from any specific initiating event, which are necessarily functions of the states and of the specific time frame, are also functions of the vulnerability and resilience of the system to the specific initiating event [Haimes 2009a]. For example, *resilience* represents *the ability of the system to withstand a disruption within acceptable degradation parameters and to recover within acceptable losses and time*. Other characterization of the risk function in terms of robustness, flexibility, and sustainability are discussed in the literature [Jugulum and Frey, 2007; de Neufville and Scholtes, 2011; Cardin et al., 2013; Hu et al., 2013].

The integration of UAS into the NAS, the transition to cloud computing, and the change in pilots' or controllers' roles are often identified as sources of vulnerabilities to the NAS. The projected likelihood of these or other vulnerabilities causing specific emergent changes should not be a roadblock to advancing the NAS, instead they should be viewed as keys to shaping the decisions needed to manage the emerging sources of risk to NextGen. Scenario structuring is a useful

tool with which to refine and assess the vulnerabilities of NextGen in the "Evolving Base" of changing conditions (to be discussed in subsequent sections). For example, in the integration of UAS into the NAS, a decision on the flight rules governing a UAS flight will affect three shared states in the NAS: (i) the controller workload; (ii) the vehicle technology requirements; and (iii) the pilot qualifications. Emergent forced changes to any one of these states will likely alter options available to stakeholders and other decision makers. Similar to vulnerability, the resilience of a system is also multidimensional and, in many ways, the vulnerability and resilience of a system are two sides of the same coin.

The current NAS emerged from a series of development efforts that did not explicitly include resilience of the air traffic management capability as a design goal. Individual systems were protected against single point failures, but the collective systems of system has no established overall plan for resilience. So the state of resilience is not assured. One of the design objectives of NextGen would be to build in resilience of the system of systems, making it less vulnerable to emergent forced changes. To avoid recreating the kind of crystal like growth that characterizes the present NAS where capabilities were added on as needed, the future must capture a more complete design of services.

2.1. Project Life Cycle and Team Risk Management

An often-neglected aspect of project risk management is attention to the entire project life cycle. Manufacturing firms, for example, commonly conduct a failure mode and effects analysis (FMEA), and failure mode, effects, and criticality analysis (FMECA) on the product and the assembly line, but they ignore the product development and design processes. Doing so ignores the risks inherent in requirements definition, development, acquisition, and phase-out or upgrade. Sage [1992, 1995] discusses the different types of risk inherent

in the various stages of the life cycle. Ignoring important stages of the life cycle can lead to substantial problems in terms of programmatic risk for both product development at the beginning of the life cycle and for product upgrade or replacement at the end. If major risks are not handled sufficiently early, they may magnify their effects later in the project. For example, in information technology acquisitions, errors in the requirements definition phase can lead to costly cascading problems later when the information system fails to meet the customer's needs. As a result, costly modifications may be necessary, causing schedule slips and cost overrun.

Considering the shifting operating environment for NextGen tends to alter the requirements. For example, an operational tool designed initially to sequence flights to maximize runway occupancy during capability constraints may be consequently required to sequence and space flights for efficiency as fuel prices rise. Versatility and adaptability to uncertain future demands, fleet mix and general operating environment (the external forced changes) are characteristics of NextGen. A case in point is the need to be cognizant of and responsive to the *Evolving Base* [Haimes 2012b]: "Dynamic shifting rules and realities: (i) Goals and objectives; (ii) Stakeholders, decision makers, and interest groups; (iii) Organizational, political, and budgetary baselines; (iv) Reorganization and reallocation of key personnel; and (v) Requirements, specifications, and delivery. For example, requirements necessarily evolve in response to external changes. If they are not adequately managed in response to the evolving base, the resulting consequences can be not only expensive, but they may not serve future needs. Similarly, if the requirements are too fluid, costs and schedules cannot be maintained. Common to all decisions, appropriate tradeoffs must be made.

Managing the risks inherent in any system is contingent upon having sufficient knowledge of the system's structure and operations. Indeed, this knowledge is imperative in order to comprehensively identify the risks associated with project acquisition and management, accurately estimate the probabilities of failure, and correctly predict the consequences of those failures. While the tendency to collect data and information on the project is important, databases are useful only when accompanied by an understanding of the way the system they describe operates. Knowledge of a system provides a means to understand and effectively use information from the system. Obtaining this knowledge is often difficult even for a single system and the problem is compounded with the development or acquisition of S-o-S. Knowledge of the many component systems is required, and it is also critical to understand the boundaries where these systems interact and generate new sources of risk. These interactions include project requirements and specifications, design and construction, finance and management, development of new technology, and response to a myriad of changes and conflicting signals from the many participating organizations. Thus, the sheer amount of system knowledge requisite for the risk analysis imposes some difficulties in its collection, dissemination, and integration.

In their book *Working Knowledge*, Davenport and Prusak [1998] suggest that knowledge moves through an organization via markets just as any other scarce resource does. There are buyers, sellers, and brokers of knowledge. Those who

possess it will sell their knowledge if properly compensated with money, reciprocity, reputé, bonuses, promotions, or other gain. If there is not sufficient compensation for those who sell their knowledge, the transfer will not take place. This market for knowledge has some important implications for risk management. The knowledge necessary to assess the risks to an entire project is spread over many individuals in multiple organizations and at multiple levels in the management hierarchy. For this knowledge to be transferred and collected for the purposes of risk management, an efficient knowledge market must exist. To this end, management and corporate culture are key influences that must facilitate rather than hinder the operation of knowledge markets. First and foremost, trust is required for the exchange of knowledge [Davenport and Prusak, 1998; Covey, 2008].

Team risk management [Haimes 2009b] brings together all of the disparate parties in the risk management effort. Katzenbach and Smith [1999] introduce the following definition: "A team is a small number of people with complementary skills who are committed to a common purpose, performance goals, and approach for which they hold themselves mutually accountable." In other words, when conducting risk management in teams, participants are imbued with a common purpose. Risk management is not externally enforced; rather, it is a process in which everyone participates. When all participants have personal stakes in the process, they are much more likely to share their system knowledge since they can see the potential benefits from doing so. To facilitate this process, organizational boundaries must be overcome. An effective means is to bring people together in face-to-face meetings: individuals from the various participating organizations, from subdivisions within organizations, and from different levels in the management hierarchy.

2.2. FAA NextGen: A Dynamic Roadmap

NextGen is comprised of S-o-S encompassing hardware and software, humans, policies and procedures, national airspace, vehicles including unmanned vehicles. It depends on other S-o-S, such as GPS, commercial telecommunications, national weather service, passengers, military, and international cooperation and partnership. Furthermore, decisions for this complex enterprise are made by a combination of individual stakeholders and organizations in order to maintain control of the states of the system under consideration to achieve specific local objectives and outcomes. To address the myriad of sources of risk associated with the emergent development of NextGen, an extension of the *Dynamic Roadmap for Risk Modeling, Planning, Assessment, Management, and Communication for Emergent S-o-S* [Haimes 2012b] is applied to NextGen with a focus on the automation portion of NextGen. The major flight data processing subsystems addressed here, which constitute a system of systems, include: (i) System-wide Information Management (SWIM); (ii) En Route Automation Modernization (ERAM); (iii) Terminal Automation Modernization and Replacement (TAMR); (iv) Collaborative Air Traffic Management Technologies (CATMT); and (v) Terminal Flight Data Manager (TFDM). The motivation for this

application is that the current path will not drive automation convergence or alignment without a systemic and principle-guided risk modeling, assessment, management, and communication process.

The FAA must sequence and manage its decision-making activities under the following constraining realities: (i) adhering to a desired *time frame*; (ii) addressing all competing *objectives* affected by the decisions; (iii) operating in the context of assumed linkage of *alternatives* to outcomes; (iv) functioning in the context of *decisions made by individual powerbrokers and stakeholders* who are striving for state changes that are beneficial to their objectives and outcomes; (v) addressing the operating modes explicitly, including *degraded modes*; (vi) addressing all questions in the risk assessment, management, and communication processes to identify the *emergent forced changes* that may alter outcomes; and (vii) recognizing *first order and indirect impacts on each subsystem*, especially those cases that pose a risk of low probability yet extreme and dire consequences.

2.3. Applying the Dynamic Roadmap and Its Extensions

All emergent S-o-S are subject to emergent forced changes. (*External or internal sources of risk to a subsystem or to the entire S-o-S that may adversely affect specific states of that system.*) Thus, the *Journey* (associated with the Dynamic Roadmap) starts with the *Evolving Base* (Dynamic shifting rules and realities: (i) goals and objectives; (ii) stakeholders, decision makers, and interest groups; (iii) organizational, political, and budgetary baselines; (iv) reorganization and reallocation of key personnel; and (v) requirements, specifications, delivery, users, and clients.); *namely, the dynamic emergent forced changes* that dominate the entire lifecycle of emergent S-o-S.

2.3.1. First Principle: Holism is the Common Denominator that Bridges Risk Analysis and Systems Engineering

The imperative adherence to the Gestalt-holistic philosophy must be the *sine qua non* for an effective modeling of the NextGen as emergent S-o-S. Consider the following sample of multiple perspectives that must be addressed: (a) the integration of the Navigation, Surveillance, and Communication systems, which currently operate semi-independently to avoid one path to failure; (b) the integration of the UAS with the NAS. To be more specific, this sample of systems integration, among numerous others, requires accounting for the corresponding subsystems; the associated shared states; decisions, decision makers, and stakeholders; policies, procedures, and regulations; integration of old with technology, retraining of technical and service personnel, and streamlining operational software with the corresponding hardware systems; harmonizing organizational culture and tradition that characterize each of the current Navigation, Surveillance, and Communication systems (which have been operating semi-independently for close to 40 years); new privacy, safety, and many heretofore unforeseen sources of risk associated with

the integration of the UAS with the NAS; and the impacts of nearly 90 decisions (structural, operational, policy, etc.) that must be made during the Alpha period, which are likely to affect and constrain the progress to be made during the subsequent Bravo and Charlie periods. These and a welter of other considerations must be addressed within the Gestalt-holistic philosophy.

2.3.2. Second Principle: The Process of Risk Modeling, Assessment, Management, and Communication must be Methodical, Disciplined, Systemic, Integrated, and Commensurate in its Comprehensiveness with the Criticality of the Systems being Addressed and of their Associated Risks

Being cognizant of the reality of the *Evolving Base* associated with NextGen must guide the risk modeling, assessment, management, and communication processes. Given the importance of risk and uncertainty analyses to NextGen, the development of which spans two decades, it will be useful to distinguish between risk analysis and uncertainty analysis, and within each, its different parts and perspectives. *Risk management* is commonly distinguished from *risk assessment*, even though some may use the term *risk management* to connote the entire process of risk assessment, management, and communication. The term *management* may vary in meaning according to the discipline involved and/or the context.

In risk assessment, the analyst often attempts to answer the following three questions [Kaplan and Garrick, 1981]: What can go wrong? What is the likelihood that it will go wrong? What are the consequences? Here we add a fourth question: "What is the time frame?" Answers to these questions help risk analysts identify, measure, quantify, and evaluate risks and their consequences and impacts.

Risk management builds on the risk assessment process by seeking answers to a second set of three questions [Haimes, 1991]: What can be done and what options are available? What are the associated tradeoffs among all relevant costs, benefits, and risks (both mitigated and residual)? What are the impacts of current management decisions on future options? This third question is the most significant for emergent the NextGen S-o-S because decisions during one time period (e.g., during the Alpha 5-year period) will affect many decisions at the subsequent Bravo and Charlie periods lasting more than 10 years combined. Risk communication must harmonize the risk assessment and management processes, as well as the communication between the risk analysts and decision makers and other stakeholders.

Effective risk communication undoubtedly plays a dominant role in harmonizing the responses to the *Evolving Base* (within and among the subsystems of the NextGen S-o-S) by all involved parties. Decision makers and other stakeholders benefit when they are able to clearly describe the challenges facing them for which they are seeking a better understanding in order to formulate appropriate policies and possible solutions. In turn, risk analysts, systems engineers, and systems integrators must be able to translate complex technical concepts, analyses, and results into language that decision makers

and other stakeholders can relate, understand, and incorporate into actionable decisions.

NextGen technical and infrastructure-based decisions, policies, regulations, and procedures are continuously being formulated and implemented. Recognizing that any and all current decisions are likely to have major impacts on future options on the emergent NextGen S-o-S, it becomes imperative to harmonize decisions made during the Alpha period with decisions to be made during the Bravo and Charlie periods to avoid structural foundational mishaps affecting current and future infrastructure systems. Between 2014 and 2017 the FAA will make numerous decisions, ranging from flight protocol standards and procedures to infrastructure design options that will dictate the course by which the U.S. airspace evolves throughout the next decade. Although some of these strategy decisions affect only a single subsystem of the NAS, a number of the decisions have broader implications for the NAS as a whole, and have a high likelihood of incurring programmatic risks (e.g., cost overruns and delays in meeting delivery schedules) if they are not made carefully. The NextGen decisions were analyzed to understand potential interactions as examples for the principles presented here.

There is a continuous need to harmonize arising conflicts among the planners and stakeholders, keeping in mind the *Evolving Base* and responsibilities during the three planning periods. In this context, it is reasonable to assume that the NextGen planners and executives, Congressional committees, U.S. Department of Homeland Security, airline carriers, pilot associations, and other stakeholders, have the following emergent issues on their agendas: (i) revisit the scope and objectives of select infrastructure systems and communicate them to the team; (ii) provide a clear insight on what would be an acceptable range of alternatives and their projected impacts on future options; (iii) collect data on each infrastructure during its construction; (iv) develop metrics with which to measure progress, and use the results to populate the *Dynamic Roadmap*; (v) continuously revisit the *Dynamic Roadmap* to ensure an adherence to its principles and guidelines; (vi) be cognizant of the fact that two-thirds of the major decisions made during the Alpha period relate to infrastructure decisions affecting navigation and surveillance systems; and (vii) assess the impact of the above on workforce, budget, schedule, cost, and the system's performance criteria.

In the multidecision-maker environment that characterizes NextGen, the perspectives and reactions of one decision maker to decisions and actions of another may alter the technical and programmatic risks for the enterprise. Consider efforts to define services offered based on better flight performance, or better performing—better served. Procedures for certain airport and weather conditions would allow flights with improved navigational performance to continue to their destination airport while those less equipped would have to be delayed or diverted. The decision for how to implement this procedure may be motivated by either a desire to incentivize aircraft operators to equip aircraft with higher levels of navigational performance, or a desire by the operators already equipped with improved performance to get the most from their investment. If the FAA procedures are designed to operate with low percentage of the fleet equipped to start, but

continue to deliver benefits even as the incentives take affect and the percentage grows then they satisfy both objectives. If they only give benefits so long as the fraction of the fleet able to participate is small, but the benefits diminish with higher levels of participation then the incentives also fade with time. Each decision maker must decide to join or not as others are weighing similar decisions.

2.3.3. Third Principle: Models and State Variables are Central to Quantitative Risk Analysis

The states of a system constitute the main building blocks of both analytical and simulation models. Models are built to answer specific questions, and must be made as simple as possible and as complex as required. Without models, risk analysts would find it nearly impossible to do their work. The centrality of state variables in decision making, and particularly in risk analysis, has not been sufficiently emphasized. Indeed, a representative model can effectively provide answers to the questions for which it is built only by first identifying and incorporating all relevant and critical states of the system. As noted earlier, the vulnerability and resilience of a system are *manifestations of the states of the system, and each constitutes a vector that is time- and threat (emergent forced changes)-dependent*.

Each decision in NextGen can be modeled as one or more state changes that represent standard elements of a concept of operations, including the expected performance, operating environment, policy constraints, operating procedures, roles and responsibilities, technology, and information quality in use by the system. To understand joint susceptibility to emergent forced changes, these state changes can be analyzed for their overlap and the magnitude of their impact on related decisions. A clustering technique can be used to group decisions with significant shared state space for further consideration regarding sequencing of decisions. This clustering was demonstrated in section A for the NextGen decisions on (i) a possible decision to create high performance airspace (segregating certain operations), (ii) a decision on automation needs to serve a more diverse fleet, and (iii) a decision on the aircraft role in separation and collision avoidance.

2.3.4. Fourth Principle: Multiple Conflicting and Competing Objectives are Inherent in Risk Management

This principle, common to all emergent S-o-S, was discussed in Section A.6, where the concept of Pareto-optimality or identifying a noninferior solution to a multiobjective optimization problem was introduced. A solution to a two-objective function is termed Pareto-optimal if improving one objective function can be achieved only by degrading the other. Indeed, Pareto-optimality is central to determining the tradeoffs between the additional expenditure required for further risk reduction.

In the NextGen S-o-S, interest in high performance airspace is tied to a gain in efficiency for advanced capability aircraft, while automation is designed to integrate a more diverse fleet and would favor access. Both may coexist at some

level, based on where performance diversity and similarities are greatest in the NAS. The third decision on avionics' role in separation and collision avoidance could support either paradigm, but it depends on knowledge of the constraints imposed by the first two.

2.3.5. Fifth Principle: Risk Analysis must Account for Epistemic and Aleatory Uncertainties

Uncertainty, commonly viewed as *the inability to determine the true state of a system*, is characterized by two sources that affect a system's modeling and thus its risk analysis [Morgan and Henrion, 1990; Paté-Cornell, 1996; Apostolakis, 1999]: (i) *Incomplete knowledge (epistemic) uncertainty*, which manifests itself in the selection of model topology (structure) and model parameters, among other sources of ignorance (e.g., lack of knowledge of important interdependencies within the system and among other systems); and (ii) *stochastic variability (aleatory) uncertainty*, which includes all relevant and important random processes and events, as well as emergent forced changes. Uncertainty dominates most decision-making processes and is the Achilles' heel for all deterministic, and most probabilistic, models. Both categories of uncertainty markedly affect the quality and effectiveness of risk analysis efforts, and ultimately, the decision-making process.

For the NextGen model, three of the states, the service policy, operational roles and personnel qualities, and the ATM technologies are more susceptible to epistemic uncertainties because the planning horizon covers multiple decades. The initial intent of policy constraints, the operating procedures, roles and responsibilities, and technology changes can be approximated, but there is little practical basis for how to model these changes over time, especially in reaction to external forced changes from the other stakeholders. The other NextGen shared states, operating environment, aircraft responsiveness, and information quality are a function of industry economics and operating practices and so are more conducive to treatment as stochastic variables with external drivers such as fuel price and market demographics. Thus, a range of future operating environments and expected performance can be used to understand the coming 20-year need for high performance airspace or divergent fleet access. The policy and operating procedures will change with time in ways that cannot be modeled; consequently the technology must be robust and adaptable enough to serve a wide range of policies and procedures. The epistemic and aleatory uncertainties identified in Figure 5 are associated with a select set of states of the NAS, noting that often most of these states are subject to both categories of uncertainties.

2.3.6. Sixth Principle: Risk Analysis must Account for Risks of Low Probability with Extreme Consequences

The most common quantification of risk is the use of the mathematical construct known as the *expected-value-of-risk* metric, which equates low probability of high-consequence events with high probability of low-consequence events. This

metric has played a decisive role in masking the criticality of extreme and catastrophic events. It is important to recognize the misuse of the expected value—the averaging of risk—when it is used as the sole criterion for risk in decision making. The partitioned multiobjective risk method (PMRM) [Asbeck and Haimes, 1984; Haimes, 2009b] supplements the expected value measure of risk with a conditional expected value of risk of extreme events. A *conditional expectation* is defined as *the expected value of a random variable, given that this value lies within some pre-specified probability or consequences range*.

In the NextGen model, risk ratings for the state changes were based on the consequence of decisions made by one stakeholder on other stakeholders. Thus, state changes to the operational roles were rated *Very High* if public/private roles, size of workforce, or harmonization between workforces of different organizations were required, regardless of their probabilities. The assumption was that with multiple stakeholders involved in emergent system planning, it is necessary to assume the most significant consequences will result from the groups' interactions and then work to mitigate them.

2.3.7. Seventh Principle: The Time Frame is Central to Quantitative Risk Analysis

Systems modeling and risk analysis cannot be performed effectively, if at all, without considering the time frame of the analysis and decision-making processes. It is important to note that the states of all systems, and thus their risk functions, are dynamic. All are functions of the time frame—even though they may be assumed, in some studies, to be static for simplicity. Furthermore, the dynamic tradeoffs among multiple objectives both complement and supplement the determination of Pareto-optimality in risk management, given the dynamic nature of evolving events. Similarly, the process of eliciting the decision makers' preferences must be systemic and harmonious with the realization that conditions, and thus policies, are dynamic and subject to being updated and modified.

2.4. Summary

There is no single method to manage the risks associated with the evolution of emergent S-o-S to achieve a set of desired outcomes. However, the principles offered in this article indicate that in meeting the goals and objectives set forth for NextGen, the FAA must sequence and manage its decision-making activities under the following constraining realities: (i) operate within a desired time frame; (ii) address all competing objectives affected by the decisions; (iii) assume linkage to outcomes associated with the alternatives; (iv) operate in the context of decisions made by individual powerbrokers and stakeholders who are striving for state changes beneficial to their objectives and outcomes; (v) address the operating modes explicitly, including degraded modes; (vi) address all questions in the risk assessment, management and communication processes to identify the emergent forced changes

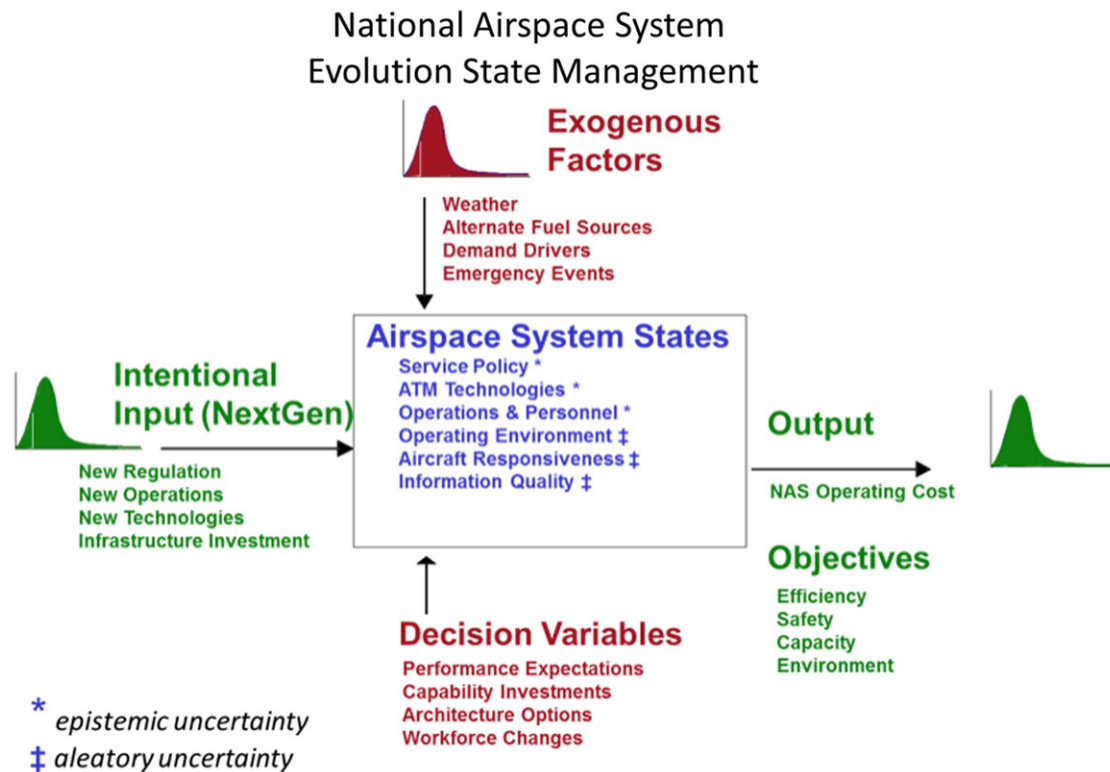


Figure 5. State space depiction of the National Airspace System.

that may alter outcomes; and (vii) recognize first order and indirect impacts on each subsystem, especially those cases that pose a risk of low probability with extreme and dire consequences. Many of the critical decisions were modeled at a high level in terms of six states (Service Policy, ATM Technology, Operations and Personnel, Operating Environment, Aircraft Responsiveness, and Information Quality) to understand how to apply this reality to the practice of continuously monitoring the Evolving Base for such complex S-o-S. The insights gained by the FAA show that by focusing on the most significant consequence of each state change it is possible to cluster decisions and work on mitigation strategies for the sequencing of decisions. Following, we recommend a more comprehensive and continuous treatment of NextGen evolution:

1. Develop criteria in order to prioritize decisions made during each period (Alpha, Bravo and Charlie) and to assess the impact of such decisions on safety, efficiency, environment, and capacity.
2. Focus on decisions where expected impacts are assessed to be high and very high and the associated consequences are deemed to be significant. To accomplish this, apply the Risk Filtering and Ranking Method (RFRM) [Haimes et al., 2002; Haimes 2009] to reduce the 90 decisions to a manageable high-priority number of decisions. Use the “Consequences” column in the RFRM framework to list the affected (i) states/constraints/objectives; (ii) decisions and inputs; and (iii) systems within NextGen. In the “Probabilities”

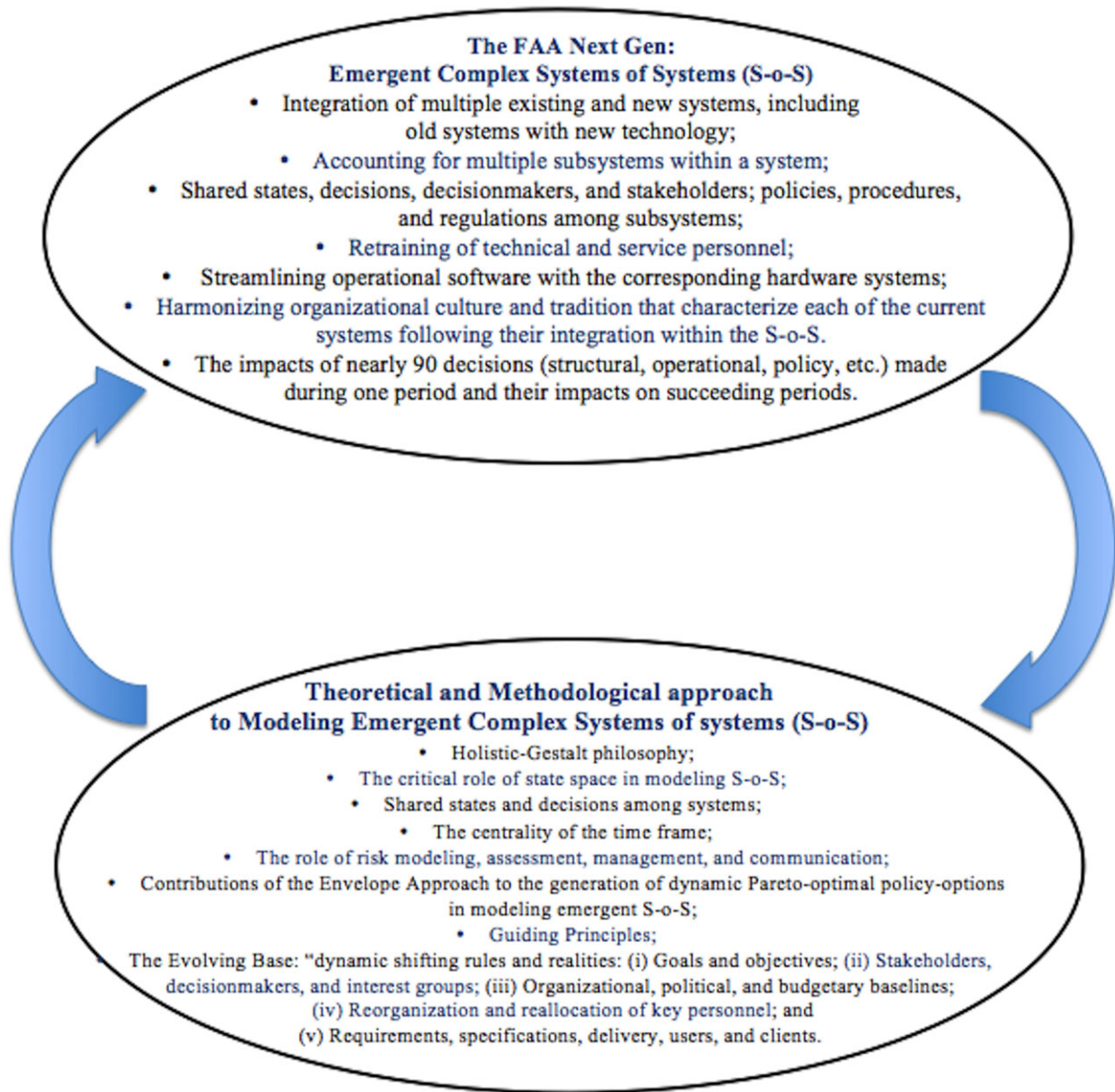
column of the RFRM, indicate “very high,” “high,” and “low” for each period Alpha, Bravo, and Charlie. As a first cut, select all entries (of the 90 decisions) that fall within very high and high (for Alpha, Bravo, and Charlie) [Werner, Haimes, and Andereg, 2014].

3. The specific probabilities required to populate this framework must be built on the collected databases, on expert evidence, and on assumptions made subject to our “learn-as-you-go” principle.

3. EPILOGUE

NextGen discussed in this article epitomes the development and evolution of emergent complex S-o-S whose completion is expected to span two decades. Simultaneous with its evolving development it is subject to a tedious process of modeling, assessing, managing, and communicating the myriad corresponding sources of risk. The Flowchart associates the S-o-S-based theory and methodology with the FAA’s NextGen complex project. Consider the following challenges facing the parties involved in this enterprise:

- i. How can we correctly estimate the life-cycle cost and the overall budget for the NextGen enterprise? Which decisions are the major cost drivers? Are they being taken in proper sequence to drive the most beneficial outcome for the objectives, the outputs, or the states of NextGen?



Flowchart. Flowchart that associates the S-o-S-based theory and methodology with the FAA's NextGen complex project.

- ii. Given that early decisions made during the Alpha time span may determine or constrain the options in ways that would drive the majority of the total cost, how can planners and decision makers ensure that by eliminating structural or policy options they are not adversely impacting or eliminating future options? For example, wrong decisions on operational changes even though they are not acquisitions can be very costly because they have direct impacts on workforce or technology cost drivers for the remaining decisions.
- iii. The FAA is also charged with making a large number of strategy decisions concerning the development of NextGen standards, procedures, and design choices in a short amount of time, many of which will not manifest

themselves until several years from now. The challenge in making each of these decisions is being able to evaluate its impact not only on the project that the decision pertains to, but also on interdependent subsystems of the NAS.

- iv. Since the FAA is also a regulatory government agency, requiring aircraft operators to comply with service policies can represent a transfer of cost to others. For example, solutions that segregate flows to adjacent airports based on performance based procedures in the flight deck may require operator investment, yet are fundamentally FAA actions/decisions on its own trade space. The formal rule making process addresses this type of joint investment decision, but may be too oner-

ous to do for each small decision. Packaging multiple decisions into a single dialogue for joint decision making requires understanding the interactions and therefore the sequence of the smaller decisions to construct a fruitful joint investment decision.

- v. The developers of the NextGen enterprise must be continuously cognizant of, and responsive to, the “evolving base” introduced earlier. These forced changes also have direct impacts on the dynamic multiple objectives and the associated tradeoffs that must be continuously addressed by the concerned decision makers and stakeholders.
- vi. The motivation for changes in any subsystem of the NextGen can be two fold: (a) the objectives of the local subsystem and (b) the larger desired emergent behavior for the system of systems as a whole. In particular, the FAA is commonly interested in the desired emergent behavior of the overall enterprise in addition to the local effect. At any time there is a desired set of tolerances for the qualities of service performance. As emergent forced changes influence the operation of the National Airspace, the desired emergent behavior may correspondingly change. For example capacity has long been the primary objective of modernization programs. The new emergent behavior was manifested by higher load factors, fewer flights, and ultimately more pressure for optimal fuel-routes. These emergent behaviors can also be geographical with different desired goals—locally and nationally. Thus, the Pareto-optimal frontier is a function of time and of the evolving base. One way to model this evolution is to observe and react to the changes in the evolving base; alternatively, is to use scenarios as precursors to forecast the shifts in frontiers and in desired behaviors.

Clearly, the above challenges constitute a daunting systems engineering task.

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